MATH2241 – Introduction to mathematical analysis (1st semester 2526)

Tutorial 6 Solution

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1. Which are equivalent to the definition of $\lim_{x\to a} f(x) = L$.

   (a) For any $\varepsilon > 0$, there is $\delta > 0$, such that $|x-a| < \delta$ implies $|f(x) - L| < \varepsilon$.

   (b) For any $\varepsilon > 0$, there is $\delta > 0$, such that $0 < |x-a| < \delta$ implies $0 < |f(x) - L| < \varepsilon$.

   (c) For any $0.001 > \varepsilon > 0$, there is $\delta > 0$, such that $0 < |x-a| < 2\delta$ implies $|f(x) - L| \leq \varepsilon$.

   (d) For any $\varepsilon > 0$, there is $\delta > 0$, such that $0 < |x-a| < \delta$ implies $|f(x) - L| < \varepsilon^2$.

   (e) For any $\varepsilon > 0$, there is $\delta > 0$, such that $0 < |x-a| < \delta$ implies $|f(x) - L| < \varepsilon + 1$.

   (f) For any $\varepsilon > 0$, there is $\delta > 0$, such that $0 < |x-a| < \delta + 1$ implies $|f(x) - L| < \varepsilon$.

   Solution.
   (c) and (d). ■

2. Use the $\varepsilon - \delta$ definition of limit to prove that $\lim_{x\to c} x^2 = c^2$ for any $c \in \mathbb{R}$.

   Solution.

$$\begin{aligned} |x^2 - c^2| &= |x+c||x-c| \\ &\leq (|x| + |c|)|x-c| = (|x-c+c| + |c|)|x-c| \\ &\leq (|x-c| + 2|c|)|x-c| \\ &\leq (1 + 2|c|)|x-c| && \text{if } |x-c| < 1 \\ &< \varepsilon && \text{if } |x-c| < \frac{\varepsilon}{1+2|c|} \end{aligned}$$

   Hence, for all $\varepsilon > 0$, take $\delta = \min\left\{1, \dfrac{\varepsilon}{1+2|c|}\right\} > 0$,

$$0 < |x-c| < \delta \implies |x^2 - c^2| < \varepsilon$$

   ■

3. Use the $\varepsilon - \delta$ definition of limit to prove that $\lim_{x\to c} \dfrac{1}{x} = \dfrac{1}{c}$ for any $c \in \mathbb{R}\backslash\{0\}$.

Solution.

$$\begin{aligned}
\left|\frac{1}{x}-\frac{1}{c}\right| &= \left|\frac{c-x}{cx}\right| \\
&< \frac{|c-x|}{|c(c/2)|} && \text{if } x \in (c/2, 3c/2) \iff |x-c| < |c|/2 \\
&< \varepsilon && \text{if } |x-c| < \varepsilon|c(c/2)|
\end{aligned}$$

Hence, for all $\varepsilon > 0$, take $\delta = \min\left\{\frac{|c|}{2}, \frac{c^2\varepsilon}{2}\right\} > 0$,

$$0 < |x-c| < \delta \implies \left|\frac{1}{x}-\frac{1}{c}\right| < \varepsilon$$

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4. Use the $\varepsilon - \delta$ definition of limit to prove that $\lim_{x\to 3} \frac{x^3-2}{x^2+6} = \frac{5}{3}$ for any $c \in \mathbb{R}$.

Solution.

$$\begin{aligned}
\left|\frac{x^3-2}{x^2+6}-\frac{5}{3}\right| &= \left|\frac{3x^3-6-5x^2-30}{3(x^2+6)}\right| = \left|\frac{(x-3)(3x^2+4x+12)}{3(x^2+6)}\right| \\
&\le \frac{|x-3|(|3x^2|+|4x|+|12|)}{3(0+6)} \\
&\le \frac{|x-3|(|3(4)^2|+|4(4)|+|12|)}{18} = \frac{76|x-3|}{18} && \text{if } x \in (2,4) \iff |x-3| < 1 \\
&< \varepsilon && \text{if } |x-3| < \frac{18\varepsilon}{76}
\end{aligned}$$

Hence, for all $\varepsilon > 0$, take $\delta = \frac{18\varepsilon}{76} > 0$,

$$0 < |x-3| < \delta \implies \left|\frac{x^3-2}{x^2+6}-\frac{5}{3}\right| < \varepsilon$$

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5. Prove the following Cauchy Criterion of function limit.

   $\lim_{x\to a} f(x)$ converges iff for any $\varepsilon > 0$, there is $\delta > 0$, such that

$$0 < |x-a| < \delta, 0 < |y-a| < \delta \implies |f(x)-f(y)| < \varepsilon. \tag{5.1}$$

Solution.

$\Longrightarrow$, Suppose $\lim_{x\to a} f(x) = L$. For any $\varepsilon > 0$, there is $\delta > 0$, such that $0 < |x-a| < \delta \implies |f(x) - L| < \frac{\varepsilon}{2}$. Then $0 < |x-a| < \delta, 0 < |y-a| < \delta \implies$

$$|f(x) - f(y)| = |(f(x) - L) - (f(y) - L)| \leq |f(x) - L| + |f(y) - L| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} < \varepsilon$$

$\Longleftarrow$ Conversely, assume $f(x)$ satisfies the Cauchy criterion. We first prove that $f(x)$ converges on any sequence satisfying $x_n \neq a$ and $\lim_{n\to\infty} x_n = a$. Then we prove that the convergence of "subsequence" $f(x_n)$ forces the convergence of the whole "sequence" $f(x)$.

To show the convergence of $f(x_n)$, we only need to show that it is a Cauchy sequence. For any $\varepsilon > 0$, there is $\delta > 0$, such that (5.1) holds. Then for this $\delta > 0$, there is $N \in \mathbb{N}$, such that

$$n > N \implies 0 < |x_n - a| < \delta,$$

where $0 < |x_n - a|$ follows from the assumption $|x - a| > 0$. Hence,

$$m, n > N \implies |f(x_m) - f(x_n)| < \varepsilon.$$

This proves that $f(x_n)$ is a Cauchy sequence and therefore converges to a limit L.

Next we prove that L is also the limit of the function at a. For any $\varepsilon > 0$, we find $\delta > 0$ such that (5.1) holds. Then we find one x_n satisfying $0 < |x_n - a| < \delta$ and $|f(x_n) - L| < \varepsilon$. Moreover, for any x satisfying $0 < |x - a| < \delta$, we may apply (5.1) to x and x_n to get $|f(x) - f(x_n)| < \varepsilon$. Therefore

$$|f(x) - L| < |f(x) - f(x_n)| + |f(x_n) - L| < 2\varepsilon.$$

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